INVENTOROS TANZANIA — TECHNICAL HANDOFF DOSSIER

STATUS: PRODUCTION READY **1,405 Automated Tests Passing (0 Failures)** • Branch: `main`

Repository: `https://github.com/Corboy/inventoros-tanzania.git`
Primary Tech Stack: Laravel 11 (PHP 8.2+), Vue 3 Composition API, Inertia.js, Tailwind CSS, SQLite/MySQL, Vite
Target Ecosystem: Tanzanian Retail, Wholesale, Hardware, Pharmacy, Supermarkets & Multi-Branch Inventory

1. Executive Summary for Incoming Engineers / AI Agents

This codebase represents the production-ready build of **Inventoros Tanzania**. The system has been fully localized to authentic commercial Tanzanian Swahili, stabilized with comprehensive automated tests, and enhanced with high-impact retail features (wholesale/retail pricing, thermal receipt printing, cash tendered and change calculations, one-tap periodic reporting, and 3-tier stock alerts).

2. Key Architectural Additions & Code Changes

Area / Module	Summary of Changes & Files Modified
Wholesale Price & Units	Added `wholesale_price` and `unit` columns to `products` table via migration `2026_09_03_232812_add_wholesale_price_and_unit_to_products_table.php`. Form requests and models updated with accessors.
POS Counter UX	In `resources/js/Pages/Orders/Create.vue`: added wholesale/retail price toggle per item, expanded quantity input to `w-16` with global spin button removal, integer parsing & blur normalization, cash shortage warnings, and `Ctrl+Enter` shortcut.
Thermal Printing	In `resources/js/Components/PosReceiptModal.vue`: 58mm/80mm ESC/POS thermal receipt modal with business header, TIN, customer info, itemized lines, change due, and thank-you footer.
Periodic Reporting	In `ReportController.php` and `SalesAnalysis.vue`: one-tap preset pills (Leo, Jana, Wiki Hii, Siku 7, Mwezi Huu, Mwezi Uliopita, Mwaka Huu, Custom), COGS calculation, Gross Profit, Margin %, cashier performance, and streaming CSV export.
Route Fixes	Corrected Ziggy route reference in `Reports/Index.vue` from `reports.lowStock` to canonical `reports.low-stock`. Corrected `WarehouseSwitcher.vue` route call to `warehouses.set-active`.
Sidebar Toggle	In `AppLayout.vue`: Desktop sidebar collapse button and `Ctrl+B` keyboard shortcut with `localStorage` persistence.

3. How to Run, Test, and Build (Exact Commands)

Run the Development Server:

```
cd "/Users/imac/nying timba/ENVENTORY/inventoros-tanzania"
php -d opcache.enable_cli=0 artisan serve --host=127.0.0.1 --port=8000
# Access via browser: http://127.0.0.1:8000
```

Compile Assets (Vite):

```
npm run build    # Production build
npm run dev      # Hot module replacement
```

Run Automated Test Suites:

```
# Full test suite (1,405 tests passing)
php artisan test

# Periodic reports tests
php artisan test tests/Feature/PeriodicReportTest.php tests/Feature/ReportControllerTest.php

# Live database audit (verifies all 36 screens against database.sqlite)
DB_DATABASE=database/database.sqlite php artisan test tests/Feature/LiveDatabaseAuditTest.php
```

4. Default Credentials & Seeded Users

Role	Name	Email	Password	Scope
Mkurugenzi (Admin)	Juma Rashidi	`admin@biashara.co.tz`	`password123`	Full system access
Meneja (Manager)	Neema Mwangi	`meneja@biashara.co.tz`	`password123`	Stock, receiving, POs
Muuzaji (Cashier)	Baraka John	`muuzaji@biashara.co.tz`	`password123`	POS sales & receipts only

5. Critical Gotchas & Architectural Rules (Must Read!)

> **1. OrderStatus is a BackedEnum:** `$order->status` is cast to `OrderStatus::class`. Never do `(string) $order->status` directly. Always use `$order->status instanceof \BackedEnum ? $order->status->value : (string) ($order->status ?? '')`.

> **2. Ambiguous Column Joins:** `users` and `orders` both have `organization_id` and timestamps. When writing raw joins, always explicitly qualify: `orders.organization_id`.

> **3. Multi-Tenancy Scoping:** All queries must filter by `organization_id`. Never expose cross-tenant data.

6. Next Steps & Extension Roadmap

- **TRA VFD Integration:** Connect to Tanzania Revenue Authority Virtual Fiscal Device API for automated QR code tax signing.

- **SMS Gateway:** Integrate Beem Africa or NextSMS to send instant customer purchase receipts via SMS.
- **Lipa Namba Push USSD:** Integrate direct mobile money callback verification.